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# Remagnetization of the Coast Range ophiolite at Stanley Mountain, California, during accretion near 10°N paleolatitude

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## ABSTRACT

**Paleomagnetic data are presented for a 50-m-thick sequence of Oxfordian to Tithonian sedimentary rocks conformably overlying Upper Jurassic pillow basalt within the Coast Range ophiolite at Stanley Mountain, California. These new data are similar in direction and polarity to previously published paleomagnetic data for the pillow basalt. The Jurassic sedimentary rocks were deposited during a mixed-polarity interval of the geomagnetic field, and uniformity of the remanent magnetization within the entire section of pillow basalt and sedimentary rocks indicates later remagnetization. Remagnetization of the Coast Range ophiolite is interpreted to have occurred during accretion to the continental margin, possibly by burial and low-temperature alteration related to this event. Similar paleolatitudes calculated for the ophiolite (11° ±3°) and for mid-Cretaceous sedimentary rocks of the Stanley Mountain terrane at Figueroa Mountain (6° ±5°) are consistent with remagnetization of the ophiolite at low paleolatitudes. Uniform-polarity directions for other remnants of ophiolite in southern California and elsewhere along the Pacific coast imply that these rocks were also overprinted, and their magnetic inclinations suggest remagnetization at low paleolatitudes as well. The Coast Range ophiolite at Stanley Mountain is thus inferred to have been remagnetized along the North American margin near 10°N paleolatitude between earliest and mid-Cretaceous time and subsequently transported northward by strike-slip faulting related to relative motions between the Farallon, Kula, Pacific, and North American plates.**

## INTRODUCTION

Remnants of Jurassic oceanic crust equivalent to the Coast Range ophiolite are exposed along the western margin of North America from the Vizcaino Peninsula and Cedros Island in Baja California (Kimbrough, 1985) to the San Juan Islands in northern Washington (Garver, 1988). These rocks are best known in the Coast Ranges of California and are generally found structurally above the Franciscan subduction-accretion complex to the west and depositionally beneath the overlying Great Valley fore-arc sequence to the east (Fig. 1). Isotopic ages for the Coast Range ophiolite span 153 to 165 Ma, and in southern California U-Pb ages are generally between 160 and 165 Ma (Hopson et al., 1981). Tectonic models proposed to explain the origins of the Coast Range ophiolite include formation at a mid-ocean ridge, in a marginal basin above an east-dipping or west-dipping subduction zone, and by intra-arc rifting (see Robertson, 1989). From the correspondence of the Mesozoic Franciscan subduction complex, Great Valley fore-arc sequence, and Klamath–Sierra Nevada magmatic arc to similar tectonic elements of a typical convergent plate boundary (Dickinson, 1970), it can be inferred that these rocks were formed together along the ancient California margin. Alternatively, these rocks could have formed in their respective tectonic settings, having been subsequently juxtaposed by arc-parallel, strike-slip faulting. Geologic data showing differences in sedimentological characteristics between coeval rocks in the Great Valley sequence and the Franciscan Complex, as well as between outliers and in situ parts of the Great Valley sequence, indicate separate points of origin for these rocks and a more complex tectonic history for the western margin of North America (e.g., Blake and Jones, 1981).

Paleomagnetic data bearing on the paleogeographic origins of the Franciscan Complex, Coast Range ophiolite, and Great Valley sequence have been difficult to obtain. Multiple alteration events and deep subaerial weathering in many cases have obscured or obliterated the primary magnetic remanence. Oceanic rocks within subduction complexes of northern California (Hagstrum, 1990) and Baja California (Hagstrum and Sedlock, 1991) apparently have been remagnetized, probably during accretion to the continental margin at low paleolatitudes. Frei and Blake (1987) found secondary magnetizations at four localities in the Coast Range ophiolite and Great Valley sequence of northern California and southwestern Oregon. Limited data for ophiolitic rocks from two other localities in northern California appear to be latitudinally concordant with North America, but no field stability tests are available for these data (Mankinen et al., 1991). Paleomagnetic data for remnants of the Coast Range ophiolite in southern California consistently indicate magnetization acquisition at low paleolatitudes (Kempner et al., 1976; Luyendyk and Hornafius, 1982; Beebe and Luyendyk, 1983), but field stability tests are also lacking, and only data from the Stanley Mountain terrane have been published (McWilliams and Howell, 1982). These data have been interpreted as concordant when compared to revised Jurassic paleopoles for North America, implying the lack of large-scale northward transport along the American margin during late Mesozoic and Cenozoic time (Butler et al., 1991). This study was undertaken, therefore, to determine whether the characteristic magnetization of the Coast Range ophiolite at

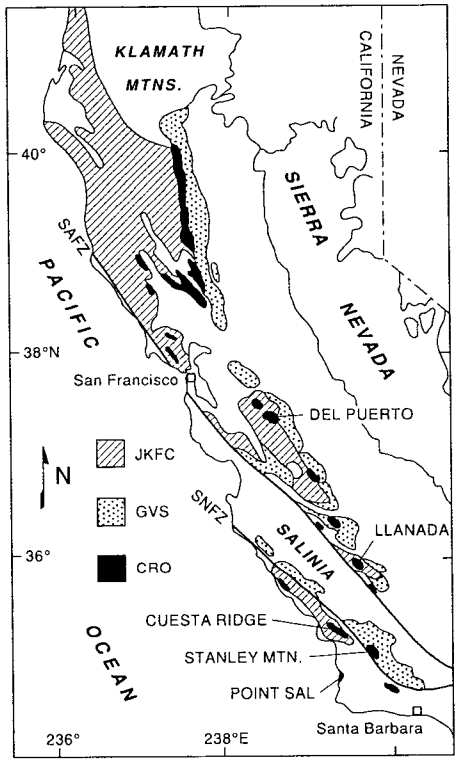



**Figure 1. Generalized tectonic map of western California (after Hopson et al., 1981) showing locations of Coast Range ophiolite (CRO) remnants discussed, specifically those in southern California. JKFC—Jurassic-Cretaceous Franciscan Complex; GVS—Great Valley sequence; SAFZ—San Andreas fault zone; SNFZ—Sur-Nacimiento fault zone.**

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Stanley Mountain is primary, giving the paleolatitude at which the ophiolite formed, or secondary, perhaps giving the paleolatitude at which the ophiolite was remagnetized, and whether this result is compatible with northward transport of the Stanley Mountain terrane.

## METHODS AND RESULTS

Samples for paleomagnetic analysis were collected from two exposures (Fig. 2) of a 50-m-thick sequence of Oxfordian to Tithonian (see Fig. 3A) tuffaceous cherts, siltstones, and limestones conformably overlying pillow basalt in Alamo Creek near Stanley Mountain, California. At site SM1, 18 core samples were drilled across the entire section at intervals ranging from 0.5 to 17 m. Six samples were taken at 0.5 to 2 m intervals over a 5-m-thick part of a second section (SM2) to the west, closer to the Huasna fault zone. Only the least altered fine-grained layers coherent enough to permit drilling were sampled. Late Miocene and younger right-lateral drag folding along the fault zone has caused site SM2 to have a significantly different structural correction than that for site SM1 (Table 1). Pillow basalts (SMB, a–j) spanning 800 m of section were sampled at 10 sites by McWilliams and Howell (1982), from just beneath the sedimentary section in Alamo Creek and continuing down section along Fish Creek (Fig. 2). Dip angles for the pillow basalts vary, but are generally around 80°NW. Strike azimuths for the pillow basalts are consistent with those for the overlying sedimentary rocks measured at site SM1, and structural corrections of the basalt data were made in this present study using the average attitude of the section at site SM1.

Alternating field (AF) demagnetization in peak inductions up to 100 mT isolated characteristic magnetizations with consistent directions, generally between 10 and 50 mT, often after removal of a recent viscous overprint (Fig. 3B). Decay of the characteristic magnetization to a point away from the origin in some samples and high-coercivity directions (>40 mT) in three tuffaceous chert samples from site SM1 with moderate negative inclinations and northeasterly declinations imply that these rocks at one time carried a third, perhaps primary, component of magnetization. Characteristic directions for samples from sites SM1 and SM2 (Fig. 3C) are well grouped. Mean directions for these two sites (Table 1) are clearly distinct, but mean directions for SM1 and for the pillow basalts (SMB) analyzed by McWilliams and Howell (1982) are indistinguishable at the 95% confidence level. After structural correction, the grouping of mean inclinations for sites SM1 and SM2 is significantly improved ($\kappa_1/\kappa_2$ = 1133; 99% confidence level), and the difference in declination between site means for SM1 and SM2 is consistent with the sense and amount of drag folding at SM2 (relative to SM1) associated with movement on the Huasna fault zone. The characteristic magnetization in these Upper Jurassic rocks, therefore, was most likely acquired prior to initiation of right-lateral movement on the fault zone in late Miocene time.

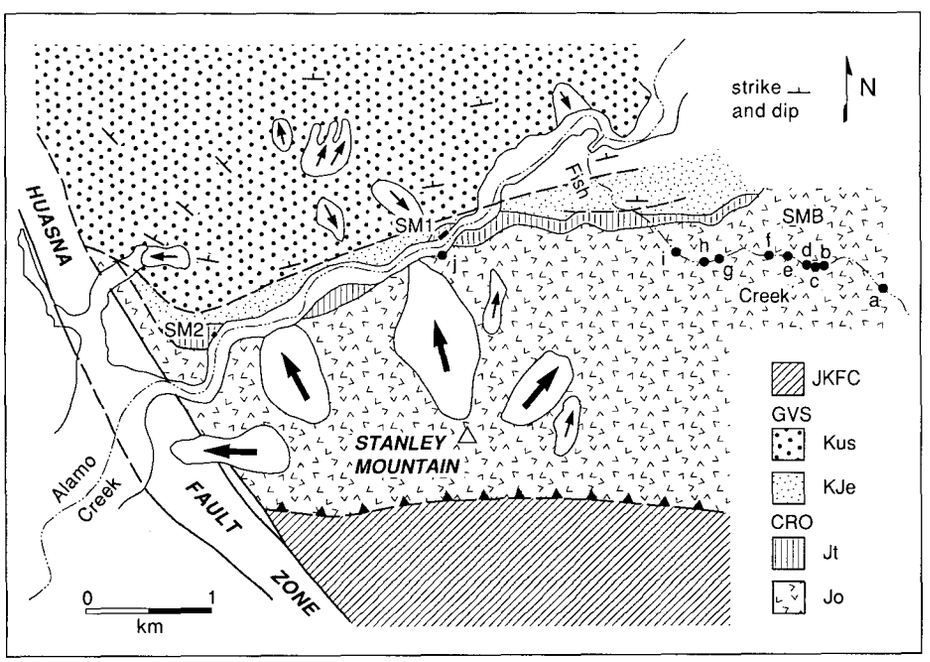



**Figure 2. Simplified geologic map of Coast Range ophiolite remnant at Stanley Mountain, California (after Vedder et al., 1989), showing paleomagnetic sampling sites from this study (SM1, SM2) and from McWilliams and Howell (1982) (SMB, a–j). Enclosed arrows indicate some landslide areas. JKFC—Jurassic-Cretaceous Franciscan Complex; GVS—Great Valley sequence; Kus—Upper Cretaceous sandstone, conglomerate, and mudstone; KJe—Cretaceous-Jurassic Espada Formation: mudstone and sandstone; CRO—Coast Range ophiolite; Jt—Jurassic tuffaceous chert, siltstone, and subordinate limestone; Jo—Jurassic spillite and keratophyre.**

## DISCUSSION AND CONCLUSIONS

The Upper Jurassic pillow basalts and overlying marine sediments of the Coast Range ophiolite exposed at Stanley Mountain were deposited during a mixed-polarity interval of the geomagnetic field (Fig. 3A), and primary directions of magnetization for these rocks would be expected to have both normal and reversed polarities. Consistent directions of the characteristic magnetization in the pillow basalt and fine-grained sedimentary rocks, however, imply remagnetization of the entire section after deposition. Remagnetization, therefore, most likely occurred between latest Jurassic and late Miocene time. Remnants of the Coast Range ophiolite, in general, have been subjected to four episodes of alteration (Hopson et al., 1981). These are (1) thermal metamorphism above the parent magma chamber, (2) hydrothermal alteration caused by circulation of heated sea water, (3) sea-floor "weathering," and (4) low-temperature alteration during accretion and burial by the Great Valley sequence at the continental margin. Remagnetization at Stanley Mountain is inferred to have occurred during accretion by chemically active fluids (e.g., Elmore and McCabe, 1991), because such alteration would affect both the pillow basalts and overlying Upper Jurassic sedimentary rocks. Remagnetization of these rocks, therefore, indicates that the paleomagnetic record bearing on the paleogeographic history of the Stanley Mountain terrane prior to accretion has been virtually obliterated.

Spatial proximity of the Stanley Mountain terrane to the continental margin is indicated by the appearance of andesite and metamorphic clasts in a Tithonian or younger conglomerate lens (Vedder et al., 1983). Continuous deposition of terrigenous sandstones (Great Valley sequence) above the ophiolite through Late Cretaceous time resulted in an overburden at least 4 km thick. The paleolatitude at which the ophiolite was accreted and remagnetized, and subsequent movements of the Stanley Mountain terrane with respect to North America, however, can be estimated only if paleohorizontal is known for the time of remagnetization. The overlying Lower to Upper Cretaceous sedimentary rocks are conformable with the ophiolite (Vedder et al., 1983, 1989) and indicate that it remained relatively flat lying during accretion and burial. Thus, if remagnetization was associated with accretion of the ophiolite during Early to mid-Cretaceous time, and the ophiolite remained flat lying, a paleolatitude of 11° ±3°N can be calculated (Table 1). McWilliams and Howell (1982) also sampled Cenomanian sedimentary rocks overlying the ophiolite at eight sites near Figueroa Mountain. Because deposition of these rocks occurred during the Creta-

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ceous normal-polarity superchron, a normal-polarity magnetization would be expected; a mean inclination of +12° gives an initial paleolatitude of 6° ±5°N, indicating that the Stanley Mountain terrane may have remained at low paleolatitudes until mid-Cretaceous time. This result could be influenced by inclination error, or possibly remagnetization, but even with a large inclination error of 15° to 20° (e.g., Anson and Kodama, 1987) these data would still indicate deposition south of 20°N paleolatitude.

Other remnants of the Jurassic Coast Range ophiolite in southern California also show shallow inclinations and low paleolatitudes after structural correction. Kempner (1977) studied the magnetic properties of the Point Sal ophiolite (Fig. 1) from the uppermost cherts to the lowermost dunites and concluded, on the basis of inclination data and estimates of structural corrections for the gabbroic sections, that the ophiolite was formed between 7° and 12°N paleolatitude. Luyendyk and Hornafius (1982) sampled pillow basalt at a site within 50 m of the overlying chert layers, and using the attitude of the overlying chert for structural correction, calculated a paleolatitude of 1° ±4°N or S for the Point Sal ophiolite. Beebe (1986) analyzed samples from five additional sites in the pillow basalts and combined these data with those of Kempner (1977) and Luyendyk and Hornafius (1982) to determine an initial paleolatitude of 2° ±3°N or S for the Point Sal ophiolite; these low paleolatitudes were thought to indicate the Jurassic paleogeographic origins of the ophiolite. However, the presence of uniform-polarity magnetizations for samples from throughout this section of oceanic crust (~3 km thick), which probably formed during a mixed-polarity interval of the geomagnetic field, implies that the Coast Range ophiolite at Point Sal also has been overprinted.

The fault-bounded Llanada ophiolite remnant is exposed in the Diablo Range east of the San Andreas fault (Fig. 1). Although the intrusive section is poorly exposed, a thick section of pillow basalts (~600 m) is well preserved and is overlain by 1200 m of andesitic and silicic volcaniclastic sedimentary rocks (Nilsen and Dibblee, 1979). Beebe (1986) sampled the top 20 m of the pillow basalt section at nine sites for paleomagnetic analysis and concluded that the Llanada ophiolite formed within 2° ±2° of the paleoequator. Paleomagnetic results for a massive lava flow and the overlying volcaniclastic rocks proved unreliable; therefore, these data were considered not to have any paleogeographic significance (Beebe, 1986). Because of its uniform-polarity characteristic magnetization, and because the inclination of this magnetization is similar to those determined for the apparently remagnetized Stanley Mountain and Point Sal remnants, rocks of the Llanada ophiolite may have been remagnetized at low paleolatitudes as well.

Beebe (1986) also sampled the Cuesta Ridge ophiolite at 11 sites west of the Nacimiento fault near San Luis Obispo (Fig. 1). The samples from only three of the 11 sites show stable uniform-polarity magnetizations and indicate, after structural correction, a paleolatitude of 9° ±33°N or S for the Cuesta Ridge ophiolite (Beebe, 1986). This result, although poor, is also consistent with remagnetization of the southern California remnants of the Coast Range ophiolite at low paleolatitudes during Cretaceous time. In addition, characteristic magnetizations for ophiolitic remnants in the Viscaino Norte terrane (Eugenia Formation) of Baja California (Hagstrum et al., 1985) and in the Decatur terrane of northern Washington (Bogue et al., 1989) show low paleolatitudes of 14° ±9° and 1° ±8°, respectively. Again, uniform-polarity magnetizations in these rocks are consistent with remagnetization. The reserved polarity of a single site in the James Island Formation of the Decatur terrane is interpreted from remagnetization circle analysis (Bogue et al., 1989) and, thus, could be an artifact of this technique. Al-

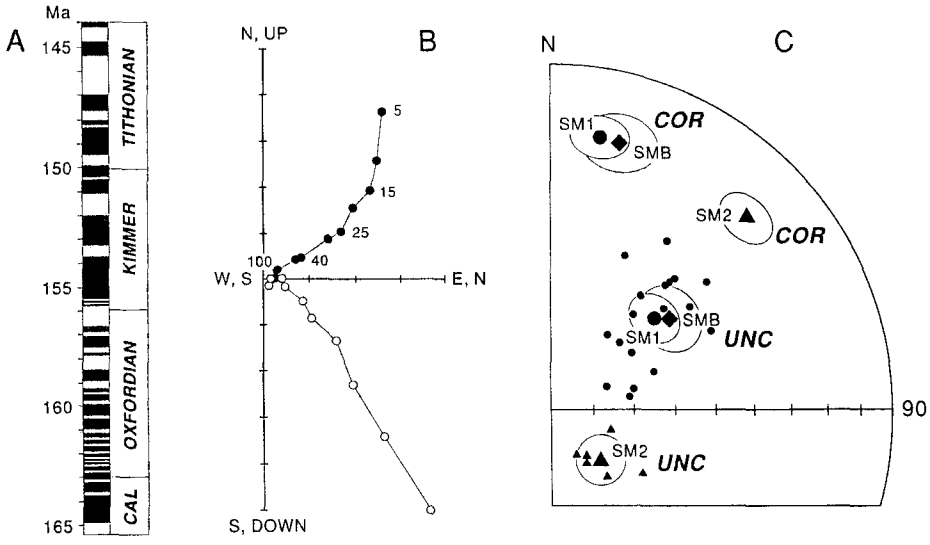



**Figure 3. A: Geomagnetic polarity time scale for Upper Jurassic after Harland et al. (1982); CAL—Callovian; KIMMER—Kimmeridgian. Black intervals indicate normal polarity. B: Orthogonal vector projection of AF demagnetization experiment for sample from site SM1; solid circles represent projection onto horizontal plane, and open circles represent projection onto north-south vertical plane. Axis ticks at 1.5 × $10^{-2}$ A·$m^{-1}$; numbers are peak AF inductions in mT. C: Equal-area projection of paleomagnetic sample data (small symbols) and site-mean directions with 95% confidence limits (large symbols) for Coast Range ophiolite at Stanley Mountain (see Table 1). *UNC*, uncorrected, and *COR*, corrected for structural tilt; symbols indicate positive inclinations.**

TABLE 1. PALEOMAGNETIC DATA FOR THE COAST RANGE OPHIOLITE AT STANLEY MOUNTAIN, CALIFORNIA

| Site | Uncorrected *I* | Uncorrected *D* | Strike/Dip | Corrected *I* | Corrected *D* | *N* | R | κ | $\alpha_{95}$ | Pole Lat | Pole Long |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Oxfordian to Tithonian sedimentary rocks | | | | | | | | | | | |
| SM1 | 57.0 | 48.9 | 247/5 | 20.9 | 10.6 | (18) | 17.5320 | 36 | 5.8 | 63.8 | 35.5 |
| SM2 | 73.5 | 133.6 | 299/73 | 20.4 | 46.1 | (6) | 5.9640 | 138 | 5.7 | 41.6 | 348.6 |
| Pillow basalts (McWilliams and Howell, 1982) | | | | | | | | | | | |
| SMB | 54.0 | 53.1 | 247/54 | 21.2 | 14.8 | 10 | 9.7646 | 38 | 7.9 | 62.4 | 27.0 |
| Averages | | | | | | | | | | | |
| SM1+B | 54.3 | 52.7 | 247/54 | 21.1 | 14.4 | 11 | 10.7625 | 42 | 7.1 | 62.5 | 27.8 |
| Inclinations only: | | | | | | | | | | | |
| All | 55.2 | | | | | 12 | | 28 | 7.1 | (37.0° ±7.1°) | |
| All | | | 247/54 | 20.9 | | 12 | | 42 | 5.8 | (10.9° ±3.1°) | |

Note: *I* and *D*, inclination and declination of paleomagnetic directions (degrees), uncorrected/corrected for structural tilt; Strike/Dip, strike and dip (dip clockwise of strike azimuth) of unit (degrees); *N*, number of sites (samples) analyzed; R, vector sum of *N* unit vectors; κ, precision parameter; $\alpha_{95}$, radius of 95% confidence (degrees); Lat and Long, latitude and east longitude of virtual geomagnetic pole corrected for tilt (degrees); Averages, average of site-mean paleomagnetic directions, average of inclinations only with paleolatitude calculated instead of virtual geomagnetic pole. Sampling locality: 35.1°N, 239.7°E.

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though remagnetization of the other ophiolite remnants at this time cannot be proven, the overall consistency of the data is compelling. Moreover, oceanic rocks within subduction complexes in northern California (Hagstrum and Murchey, 1991), central Japan (Shibuya and Sasajima, 1986), and Baja California (Hagstrum and Sedlock, 1991) also were likely remagnetized at low temperatures by chemically active fluids during accretion, implying that remagnetization is a common process within accreted oceanic rocks around the Pacific margin.

Remagnetization of the Coast Range ophiolite at Stanley Mountain and elsewhere in southern California at low paleolatitudes (~10°N) would require significant northward coastwise transport of these rocks with respect to cratonic North America since Early Cretaceous time. Northward displacements of ophiolite remnants in the Diablo Range of California also would require a major suture between these rocks and subsurface Sierran basement rocks beneath the Great Valley. A possible boundary has been identified by Holbrook and Mooney (1987) and Park et al. (1991) using seismic refraction and magnetotelluric methods, respectively. A minimum northward shift between about 20° and 35°, or 2100 to 3900 km, is needed to move the Stanley Mountain terrane from within 10° of the paleoequator to its present position in southern California, depending on the timing of remagnetization and relative position of North America between earliest and mid-Cretaceous time (150–100 Ma; see Gordon et al., 1984). Plate-reconstruction models indicate oblique subduction of the Farallon and Kula plates with respect to North America between mid-Cretaceous and early Tertiary time (Engebretson et al., 1985), and studies of modern subduction zones show that arc-parallel strike-slip faulting and tangential displacements of crustal slivers between the trench and strike-slip fault are a likely consequence of oblique convergence (Jarrard, 1986). Although ~2100 km of coastwise transport (accretion at 150 Ma) could have been accomplished by relative motions between the Farallon, Kula, Pacific, and North American plates, >3900 km (accretion at 100 Ma) is more difficult to explain. It would have required most of the available northward component of the convergence of the Farallon and Kula plates, perhaps in part as transform motion, and possibly a more northward trending margin as well (see Debiche et al., 1987). The available paleomagnetic data for the Stanley Mountain terrane of southern California, therefore, do not appear to be concordant with North America, as suggested by Butler et al. (1991), but instead appear to be consistent with the larger paleomagnetic data set for westernmost North America (e.g., Beck, 1989) indicating large northward displacements of crustal slivers along the Pacific margin during Late Cretaceous and early Tertiary time.

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